## 1. OBJECTIVE

To determine the presence of viruses in shrimp.

## 2. SCOPE

This instruction is applicable in the determination of White Spot Syndrome Virus (WSSV) and Acute Hepatopancreatic Necrosis Disease (AHPND), also known as Early Mortality Syndrome (EMS) in shrimp cultured in different prawn farms.

## 3. RESPONSIBILITY

The analyst performing the test should ensure that the procedure is followed accordingly.

## 4. APPARATUS/EQUIPMENT/CONSUMABLES

- 4.1 IQ Plus™WSSV Kit with POCKET System
  - 4.1.1 WSSV Premix Pack
  - 4.1.2 Premix Puffer B
  - 4.1.3 WSSV P(+) Control
  - 4.1.4 P(+) Control Buffer
  - 4.1.5 Inoculating Loops
- 4.2 IQ Plus™AHPND/EMS Plasmid Kit
  - 4.2.1 AHPND/EMS Plasmid Premix Pack
  - 4.2.2 Premix Buffer B
  - 4.2.3 AHPND/EMS PlasmidP(+) Control
  - 4.2.4 P(+) Control Buffer
- 4.3 IQ Plus™AHPND/EMS Toxin 1 Kit
  - 4.3.1 AHPND/EMS Toxin 1 Premix Pack
  - 4.3.2 Premix Buffer B
  - 4.3.3 AHPND/EMS Toxin 1 P(+) Control
  - 4.3.4 P(+) Control Buffer
- 4.4 Alcohol (95% Solution Ethanol)
- 4.5 IQ Plus™ Extraction Kit
  - 4.5.1 Solution 1
  - 4.5.2 Solution 2
  - 4.5.3 Solution 3
  - 4.5.4 Spin Column & Collection Tube

4.5.5. Grinder
4.6 cubee™Mini-Centrifuge
4.7 Micropipette and filter tips
4.8 Micro-Centrifuge Tubes
4.9 R-Tube
4.10 POCKIT™ Nucleic Acid Analyzer

**5. REAGENTS**

5.1 -

5.2 WSSV P(+) Control

Re-dissolve with 100µl P(+) Control Buffer before use. Store at 4°C.

5.3 AHPND/EMS Plasmid P(+) Control

Re-dissolve with 100µl P(+) Control Buffer before use. Store at 4°C.

5.4 AHPND/EMS Toxin 1 P(+) Control

Re-dissolve with 100µl P(+) Control Buffer before use. Store at 4°C.

5.5 **IQ Plus™**Extraction Kit

In the 12 ml bottle of Solution 2 add 48 ml 95% Ethanol before use.

**6. PROCEDURE**

❖ **WSSV and AHPND/EMS Plasmid Extraction Procedure**

6.1 Prepare sample according to sample type.

| Sample Type: | Sample Size: |
|---|---|
| Stomach (shrimp tissue) | 1 piece |
| Mid-gut (shrimp tissue) | 1 cm |
| <PL6 | 25-50 PLs |
| PL6-15 | 10-30 PLs |

6.1.1 Place sample into a clean 1.5ml microcentrifuge tube with 500µl Solution 1.

6.1.2 Use grinder to homogenize sample inside the tube.
6.1.3 Add 500µl Solution 2 (with ethanol) into the tube.
6.1.4Mix thoroughly and spin for 1 minute.
6.1.5 Transfer 500µl supernatant to spin column and collection tube set.
6.1.6Spin for 1 minute and discard the flow-through from the collection tube. Reassemble the collection tube to the spin column.
6.1.7 Add 500µl Solution 2 into the spin column and collection tube set.
6.1.8 Spin for 3 minutes and discard the flow-through.
6.1.9 Transfer the spin column to a clean 1.5ml microcentrifuge tube.
6.1.10 Add 200µl Solution 3into the spin column. For Hepatopancreassample, add 400µl Solution 3 into the spin column. Note that the volume of Solution 3 is subject to change according to the sample type.
6.1.11 Spin for 1 minute. The nucleic acid extract is eluted into the 1.5ml microcentrifuge tube and is ready to use.

❖ **WSSV and AHPND/EMS Plasmid Reaction Procedure**

6.2 Before preparing the reactions for iiPCR testing, turn on **POCKIT™**Nucleic Acid Analyzer to initiate the calibration for the instrument. The device will complete self-test within 5 minutes.
6.2.1 Label R-tube(s) in the labelling area.
6.2.2 Prepare one premix for each sample. The Premix tubes are in Premix Pack.
6.2.3Add 50µl Premix B to each Premix tube.
6.2.4 Use inoculating loop, take nucleic acid extracts or dissolve P(+) control into each Premix tube. Spin Premix tubes briefly in a mini centrifuge (such as cubee™Mini-Centrifuge). Note that the inoculating loop must be repeatedly dip (3 times) into the solution to collect the correct solution volume.
6.2.5Transfer 50µl Premix/ sample mixture into R-tube.
6.2.6 Seal top of each R-tube with a cap. Make sure R-tube is capped tightly.
6.2.7Place R-tube into the holder of **POCKIT™** Nucleic Acid Analyzer.
6.2.8Spin tube/holder set briefly in **cubee™**Mini-Centrifuge to make sure all solution is collected at the bottom of R-tube. Make sure that there are no bubbles in the solution. Note that the reaction must start within 1 hour to prevent nucleic acid degradation and non-specific reaction.
6.2.9**POCKIT™**Nucleic Acid Analyzer Reaction. Select “520 nm + 550 nm”. And when “System READY” is displayed, place the holder with R- tube(s) into the reaction chamber. Then tap cap of each R-tube to make sure the tube is positioned properly.

6.2.10 Close lid and press "Run" to start reaction program.

6.2.11 Test results are shown on the monitor after reaction is completed.

## 7. DATA INTERPRETATION

7.1 Signal interpretation.

Signal: 520 nm and 550 nm. 520 nm fluorescent signal is used to detect nucleic acid sequence of virus; 550 nm signal as the internal control (IC) is used to target a house keeping gene of Penaeid shrimps.

7.2 Sample results.

| 520 nm | 550 nm | Interpretation |
|---|---|---|
| + | + | Positive |
| + | - | Positive |
| - | + | Negative |
| - | - | Repeat the test. |
| + | ? | Positive |
| ? | + | Repeat reaction with freshly prepared nucleic acid. |
| - | ? | Repeat reaction with freshly prepared nucleic acid. |
| ? | - | Repeat reaction with freshly prepared nucleic acid. |
| ? | ? | Repeat reaction with freshly prepared nucleic acid. |

Note: The result showing AHPND-associated plasmid ***positive***means that the sample may containthe AHPND/EMS-associated plasmid with or without the toxin 1 virulence factor. A supplemental PCR diagnosis assay targeting toxin 1 sequence (AHPND-EMS Toxin 1) must be done. For Extraction and Reaction follow Procedures 6.1 (6.1.1-6.1.11) and 6.2. (6.2.1-6.2.11)

## 8. REFERENCES


Lightner, D.V. (ed) (1996) A handbook of pathology and diagnostic procedures for diseases of Penaeid shrimp. World Aquaculture Soc., Baton Rouge. Section 3.11

Chou, P.H. et al. (2011). Real-time target-specific detection of loop-mediated isothermal amplification for white spot syndrome virus using fluorescence energy-based probes, J. Virol. Methods 173:67-74


## 9. FORMS

LF 08-01-06, Report of Test

LF W01-FIS-01, Analyst Worksheet for Molecular Diagnosis for Shrimp